

NileCAM81_CUOAGX

Getting Started Manual



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e-con Systems

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Disclaimer

The specifications of NileCAM81_CUOAGX board and instructions on how to use this board with Jetson AGX Orin™/Xavier™ development kit is provided as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to NileCAM81_CUOAGX

NileCAM81_CUOAGX is a family of NileCAM camera module from e-con Systems. It is provided with a high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81_CUOAGX offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81_CUOAGX fast (low latency), flexible and yet reliable.

NileCAM81_CUOAGX uses Maxim's Gigabit Multimedia Serial Link (GMSL) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81_CUOAGX embeds the Maxim's serializer which serializes the high-speed image data and transmits over coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81_CUOAGX serializer recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81_CUOAGX board has been designed and developed by e-con Systems for the NVIDIA® Jetson AGX Orin™/Xavier™ development kit. NileCAM81_CUOAGX is an 8 MP GMSL camera contains a 1/1.7" AR0821 CMOS image sensor from onsemi™ along with on-board ISP. The coaxial cable is connected to camera module enclosure through a rugged FAKRA connector. NileCAM81_CUOAGX is interfaced to the J509 connector on the Orin™/Xavier™ using the NileCAM81_CUOAGX deserializer board.

The NVIDIA® Jetson Orin™/Xavier™ developer kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope.

e-con Systems provides the sample applications that demonstrate the features of this camera. However, this camera can also be utilized by any V4L2 application.

This document describes how to interface NileCAM81_CUOAGX board on Jetson Orin™/Xavier™ development kit and to check the image preview.

Prerequisites

The prerequisites are as follows:

- A Host PC running Ubuntu 20.04 Operating System
- Necessary accessories such as keyboard, mouse, HDMI monitor and so on, are required for connecting with NVIDIA® Jetson Orin™/Xavier™ development kit
- A Jetson Orin™/Xavier™ development kit pre-flashed with L4T35.4.1

Parts Supplied

The following table lists the parts supplied with the kit.

Parts Supplied	Image	Quantity
Camera module board and Serializer board		X
Deserializer board		1
Coaxial cable (3m/15m length)		X
12V 2A power adapter (For powering deserializer board)		1
Screws (deserializer to Orin kit)		3

Table 1: Parts Supplied and its Quantity

Note: Quantity X depends on the part number ordered as listed in below table.

Part Number	X
NileCAM81_CUOAGX_1H01R1 (3m Coaxial Cable)	1
NileCAM81_CUOAGX_1H02R1 (15m Coaxial Cable)	1
NileCAM81_CUOAGX_4H01R1 (3m Coaxial Cable)	4
NileCAM81_CUOAGX_4H02R1 (15m Coaxial Cable)	4
NileCAM81_CUOAGX_6H01R1 (3m Coaxial Cable)	6
NileCAM81_CUOAGX_6H02R1 (15m Coaxial Cable)	6

Table 2: Parts Numbers and its Quantity

Description

The NileCAM81_CUOAGX board is designed to interface one or six 8MP camera modules with the Jetson Orin™/Xavier™ development kit.

NileCAM81_CUOAGX is a multi-board solution, which has three boards as follows:

- e-CAM82_CUMI0821_MOD_H02R1 (Camera module)
- e-CAM22_CUMI_SER (Serializer Board)
- e-CAM_CUOAGX_DESER_6H01R1 (Deserializer Board)

The front and rear views of NileCAM81 camera module and deserializer boards are shown in following figures.

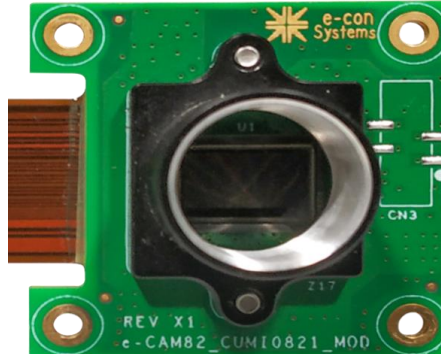


Figure 1: Front View of NileCAM81 Camera Board



Figure 2: Rear View of NileCAM81 Serializer Board



Figure 3: Front view of NileCAM81_CUOAGX Deserializer Board

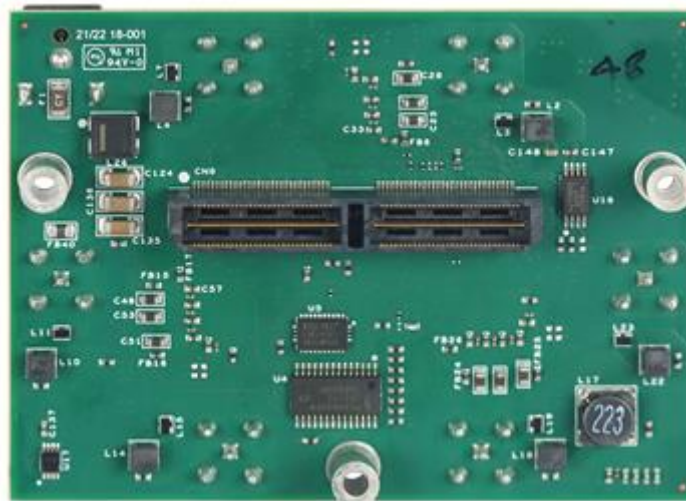


Figure 4: Rear view of NileCAM81_CUOAGX Deserializer Board

The camera board is based on AR0821, 1/1.7" CMOS image sensor from onsemi™ along with ISP. The camera is available with M12 lens holder allowing customers to choose the lens as per their requirements. The serializer board is based on Maxim GMSL serializer and has all the circuitry required for bidirectional data communication, power and data separation and so on. FAKRA connectors are used for the coaxial cable interface. The deserializer board needs external power supply, which supplies the power to the Serializer board over coaxial cable using FAKRA connector. The deserializer board has six FAKRA connectors, through which you can connect six cameras at the same time.

NileCAM81_CUOAGX is a three-board solution. The camera module has two boards, module board and serializer board, each of size as 30 mm x 30 mm. The dimension of deserializer board is 75 mm x 55 mm. NileCAM81_CUOAGX is available with both 3m and 15m coaxial cable options for power over coax applications.

Board Handling Procedure

This section describes how to connect NileCAM81_CUOAGX to the Jetson Orin™/Xavier™ developer kit. The steps required setting up NileCAM81_CUOAGX board with Jetson Orin™/Xavier™ developer kit is explained below.

- [Connecting NileCAM81_CUOAGX with Jetson Orin™ Developer Kit](#)
- [Connecting NileCAM81_CUOAGX with Jetson Xavier™ Developer Kit](#)

Connecting NileCAM81_CUOAGX with Jetson Orin Developer Kit

The NileCAM81_CUOAGX deserializer board must be mounted to J509 camera interface connector of Orin™ development kit as follows:

1. Insert the pin 1 of CN9 connector of NileCAM81_CUOAGX deserializer with the pin 1 of J509 connector of Orin™ development kit.

The NileCAM81_CUOAGX base board location in Jetson AGX Orin™ development kit is shown in following figure.

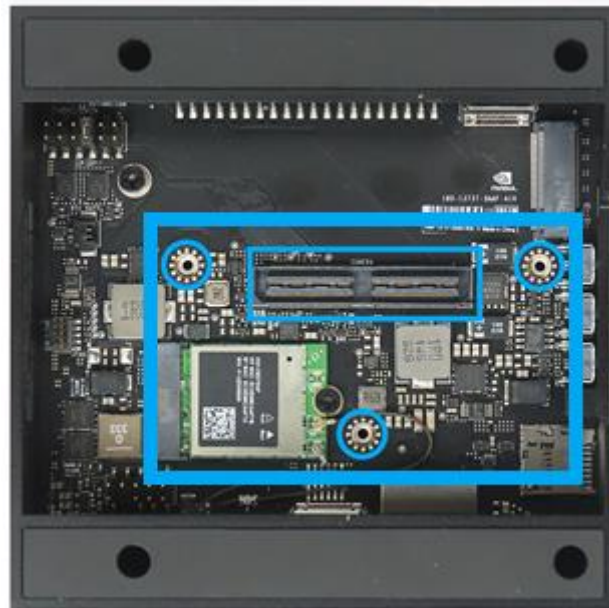


Figure 5: NileCAM81_CUOAGX Base Board Location in Jetson AGX Orin Development Kit

Note: Refer the below image for pin 1 marking between deserializer board and Jetson AGX Orin™ kit before mating.

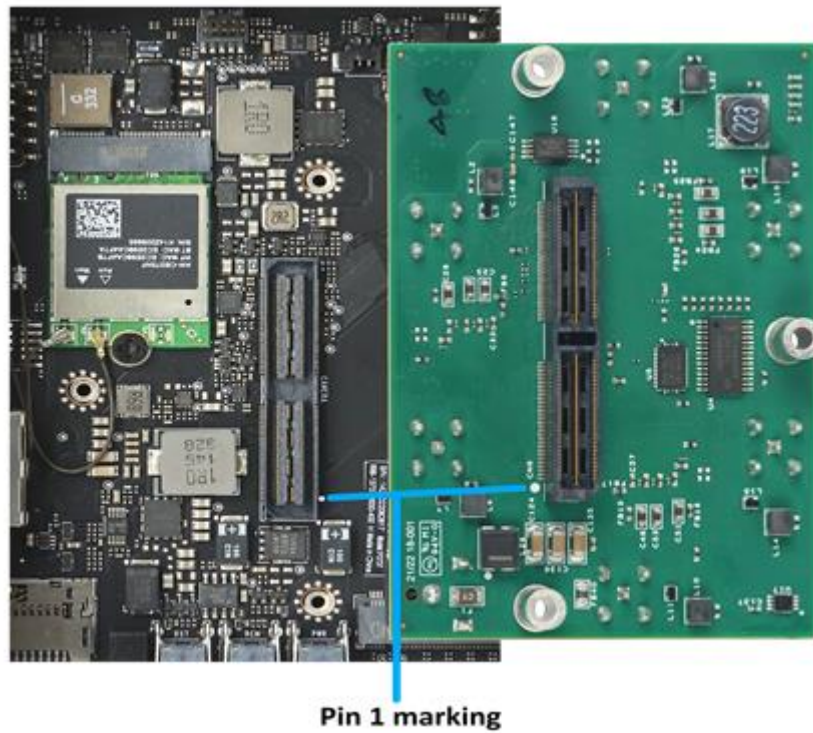


Figure 6: Location of Pins on Deserializer Board and Jetson Orin Development Kit

NileCAM81_CUOAGX deserializer board is connected to J509 connector of Orin™ development kit as shown in the following figure.



Figure 7: Deserializer connected with Jetson AGX Orin Development Kit

2. Use the provided M2.5 screws for mounting NileCAM81_CUOAGX deserializer board with Orin™ development kit for mechanical stability as shown in Figure 7.

Deserializer board has an **SW1** switch for selecting GMSL mode.

3. Select GMSL2 6Gbps mode by selecting **1 0** for A & B position. The location of SW1 is shown in the following figure.

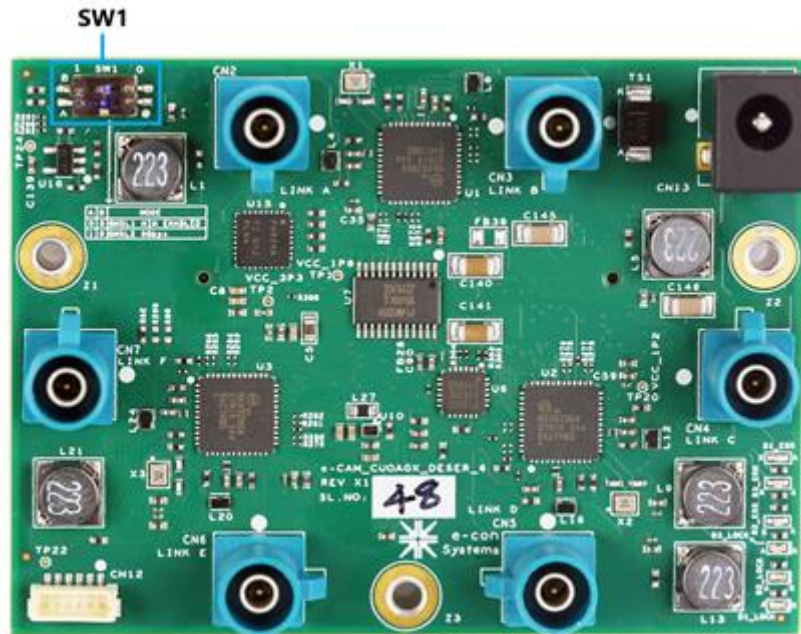


Figure 8: Position of SW1 Switch

The following table describes the different modes in the switch.

A	B	Mode
0	0	GMSL 1 HIM Enabled
0	1	GMSL 2 3Gbps
1	0	GMSL 2 6Gbps (default)
1	1	GMSL 1 HIM Disabled

Table 3: Modes in Switch

4. Identify the FAKRA connector on board.

Both the serializer and deserializer boards are provided with high frequency FAKRA straight plug-in water-blue color. The FAKRA connectors are neutrally coded, enabling FAKRA jacks of variable coding to be used in coaxial cable, if required.

The following figure shows the FAKRA plug on serializer and deserializer boards.

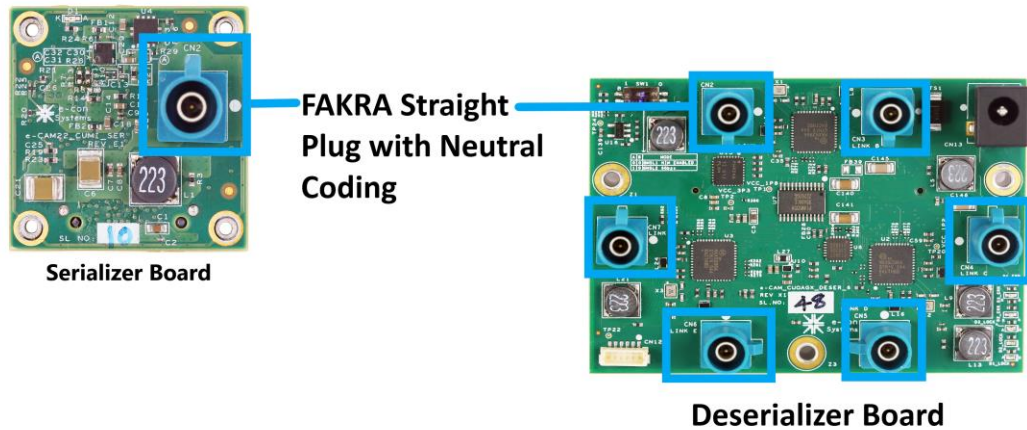


Figure 9: FAKRA Plug on Serializer and Deserializer Boards

5. Insert the coaxial cable to FAKRA connector.

The coaxial cable is provided along with the kit. The cable has a neutrally coded FAKRA straight jack in water-blue color on either end, to be inserted into the FAKRA plug on the serializer/deserializer boards.

6. Insert the FAKRA jack of the coaxial cable over the plug, such that the jack slides over the mechanical key of the plug.
7. Push the cable such that it locks properly with jack on board.

The following figure shows the cable inserted to the FAKRA plug.

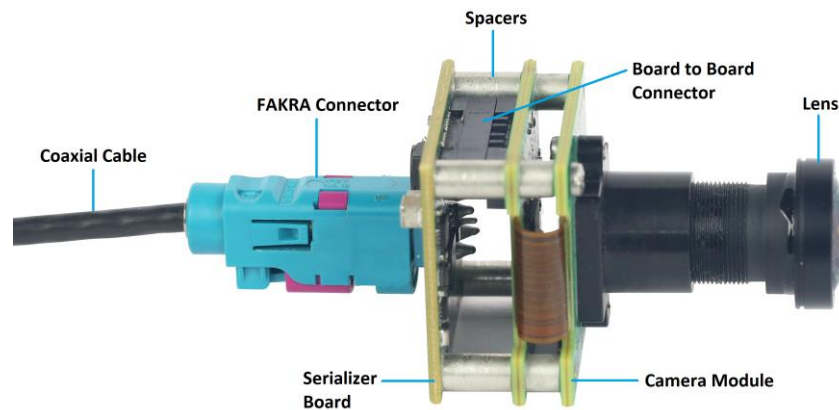


Figure 10: Inserting Coaxial Cable to FAKRA Plug of NileCAM81 Camera Module

Note:

- Any end of the cable can be inserted in any order to the plugs of serializer and deserializer boards.
- For more details in removing coaxial cable from FAKRA plug, refer to *FAQ* section.

The following figure shows the coaxial cable connected to deserializer boards.

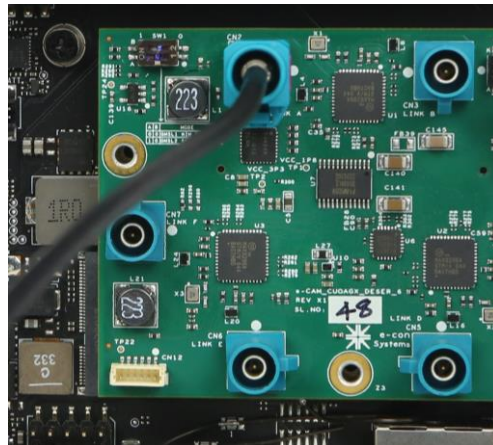


Figure 11: Inserting Coaxial Cable to FAKRA Plugs of NileCAM81_CUOAGX Deserializer

Note: The actual coaxial cable may differ from cable shown in the image.

With NileCAM81_CUOAGX connected to Jetson Orin™ development kit, follow the below steps.

8. Connect the e-con provided 12V 2A power supply to DC jack of NileCAM81_CUOAGX deserializer board for powering camera modules.

The location of DC jack on NileCAM81_CUOAGX deserializer board is shown in following figure.

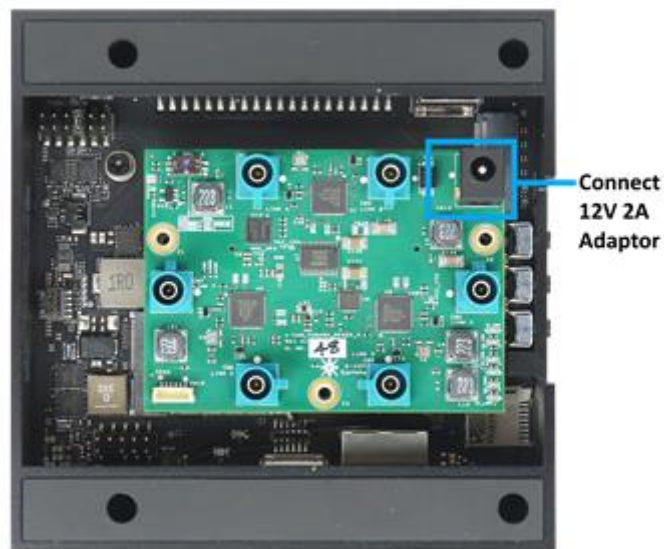


Figure 12: DC Jack Connector location in NileCAM81_CUOAGX Deserializer Board

The power supply connected to DC jack of NileCAM81_CUOAGX deserializer board is shown in following figure.



Figure 13: 12V Power Adapter connected to DC Jack of NileCAM81_CUOAGX Deserializer Board

With NileCAM81_CUOAGX connected to Jetson AGX Orin™ development kit, the steps to power up and configure the Orin™ setup are as follows:

9. Connect 20V, 4.5A supply to DC jack to power ON the Jetson Orin™ development kit.

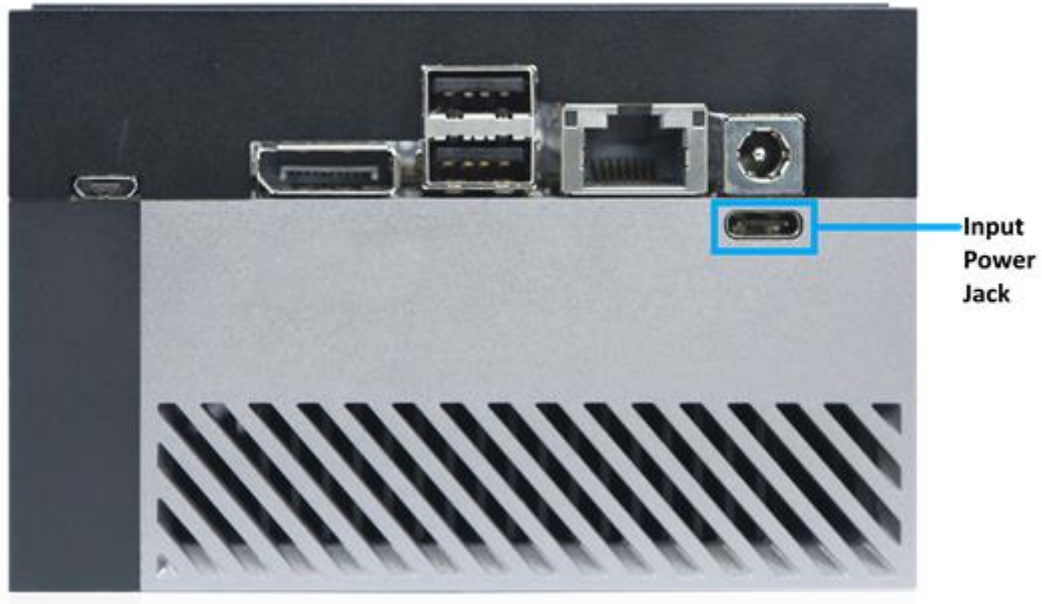


Figure 14: Location of Power Jack on Jetson AGX Orin Development Kit



Figure 15: Power Adaptor connected to Power Jack on Jetson AGX Orin Development Kit

10. Press the power switch of Jetson development kit to boot the device.

The location of power switch on the Jetson AGX Orin™ development kit is shown in following figure.



Figure 16: Location of Power Switch on Jetson AGX Orin Development Kit

After powering ON the Jetson AGX Orin™ development kit, the white color LED on Jetson AGX Orin™ development kit will glow. This serves as an indication for power-up of base board as shown in following figure.



Figure 17: Power ON Indication White LED glowing on Jetson AGX Orin Development Kit

Connecting NileCAM81_CUOAGX with Jetson Xavier Developer Kit

The NileCAM81_CUOAGX deserializer board must be mounted to J509 camera interface connector of Xavier™ development kit as follows:

1. Insert the pin 1 of CN9 connector of NileCAM81_CUOAGX deserializer with pin 1 of J509 connector of Xavier™ development kit.

The NileCAM81_CUOAGX base board location in Jetson AGX Xavier™ development kit is shown in following figure.

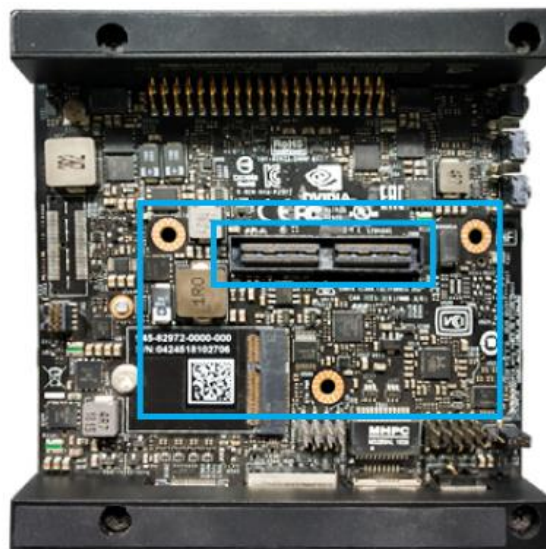


Figure 18: NileCAM81_CUOAGX Base Board Location in Jetson Xavier Development Kit

Note: Refer the below image for pin 1 marking between deserializer board and Jetson AGX Xavier™ kit before mating.

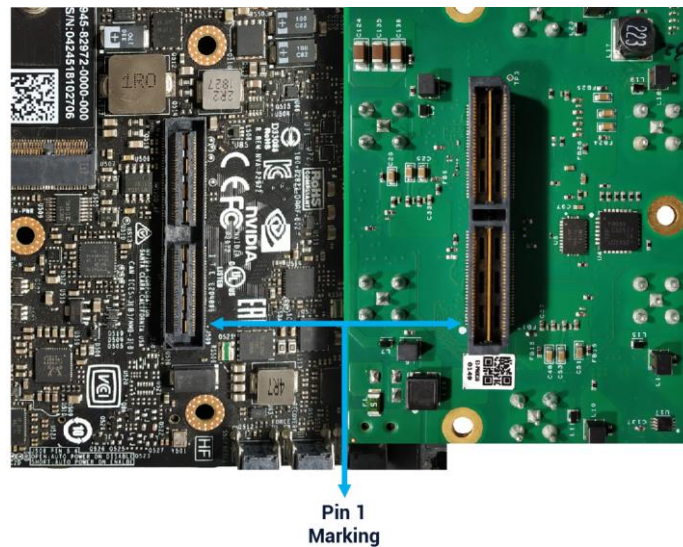


Figure 19: Location of Pins on Deserializer Board and Jetson Xavier Development Kit

NileCAM81_CUOAGX deserializer board connected with J509 connector of Xavier™ development kit.



Figure 20: Deserializer connected with Jetson AGX Xavier Development Kit

Deserializer board has an SW1 switch for selecting GMSL mode.

2. Select GMSL2 6Gbps mode by selecting **1 0** for A & B position. The location of SW1 is shown below.

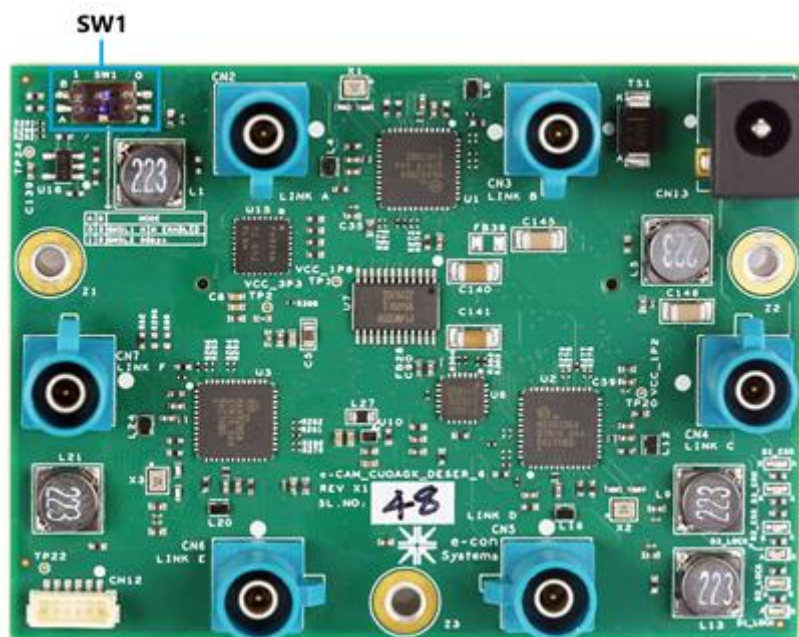


Figure 21: Position of SW1 Switch

The following table describes the different modes in the switch.

A	B	Mode
0	0	GMSL 1 HIM Enabled
0	1	GMSL 2 3Gbps
1	0	GMSL 2 6Gbps (default)
1	1	GMSL 1 HIM Disabled

Table 4: Modes in Switch

3. Identify the FAKRA connector on board.

Both the serializer and deserializer boards are provided with high frequency FAKRA straight plug-in water-blue color. The FAKRA connectors are neutrally coded, enabling FAKRA jacks of variable coding to be used in coaxial cable, if required.

The following figure shows the FAKRA plug on serializer and deserializer boards.

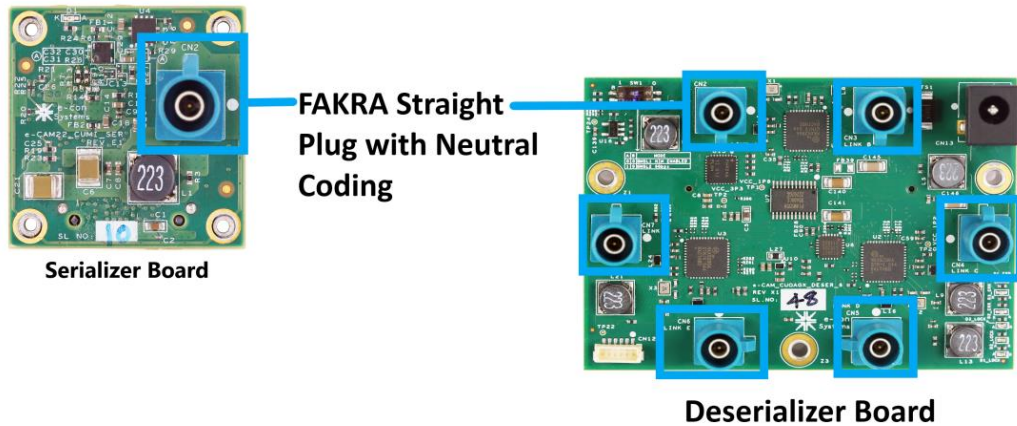


Figure 22: FAKRA Plug on Serializer and Deserializer Boards

4. Insert the coaxial cable to FAKRA connector.

The coaxial cable is provided along with the kit. The cable has a neutrally coded FAKRA straight jack in water-blue color on either end, to be inserted into the FAKRA plug on the serializer/deserializer boards.

5. Insert the FAKRA jack of the coaxial cable over the plug, such that the jack slides over the mechanical key of the plug.
6. Push the cable such that it locks properly with jack on board.

The following figure shows the cable inserted to the FAKRA plug.

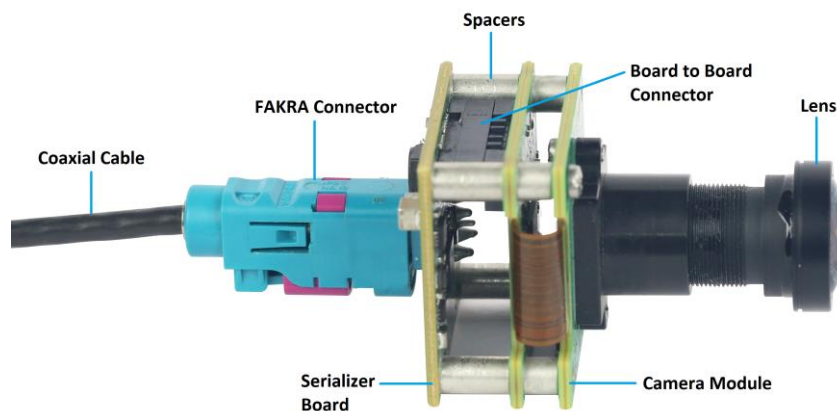


Figure 23: Inserting Coaxial Cable to FAKRA Plug of NileCAM81 Camera Module

Note:

- Any end of the cable can be inserted in any order to the plugs of serializer and deserializer boards.
- For more details in removing coaxial cable from FAKRA plug, refer to *FAQ* section.

The following figure shows the coaxial cable connected to deserializer boards.



Figure 24: Inserting Coaxial Cable to FAKRA Plugs of NileCAM81_CUOAGX Deserializer

Note: The actual coaxial cable may differ from cable shown in the image.

With NileCAM81_CUOAGX connected to Jetson Xavier™ development kit, follow below steps.

7. Connect the e-con provided 12V 2A power supply to DC jack of NileCAM81_CUOAGX deserializer board for powering camera modules.

The location of DC jack on NileCAM81_CUOAGX deserializer board is shown in following figure.

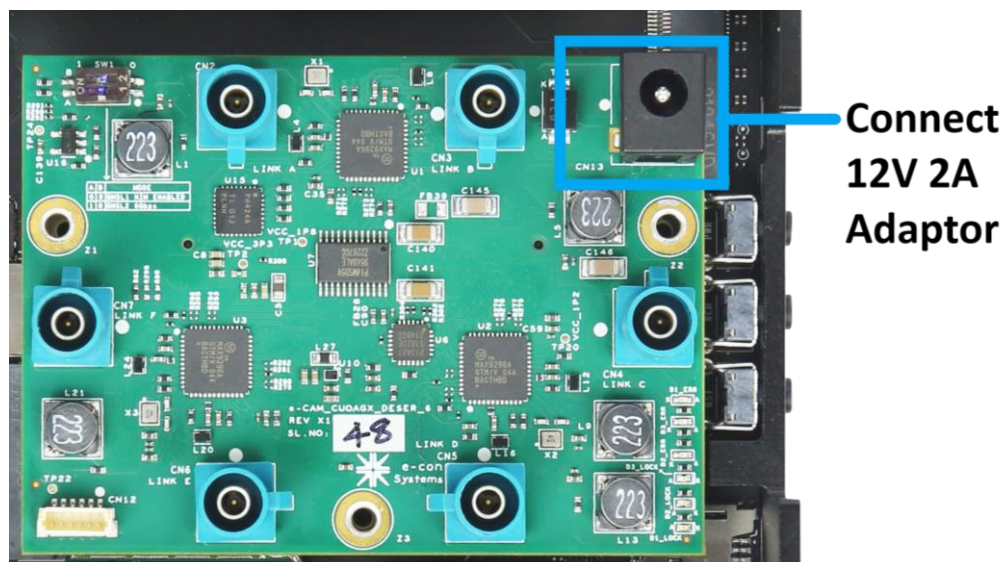


Figure 25: DC Jack Connector location in NileCAM81_CUOAGX Deserializer Board

The power supply connected to DC jack of NileCAM81_CUOAGX deserializer board is shown in following figure.

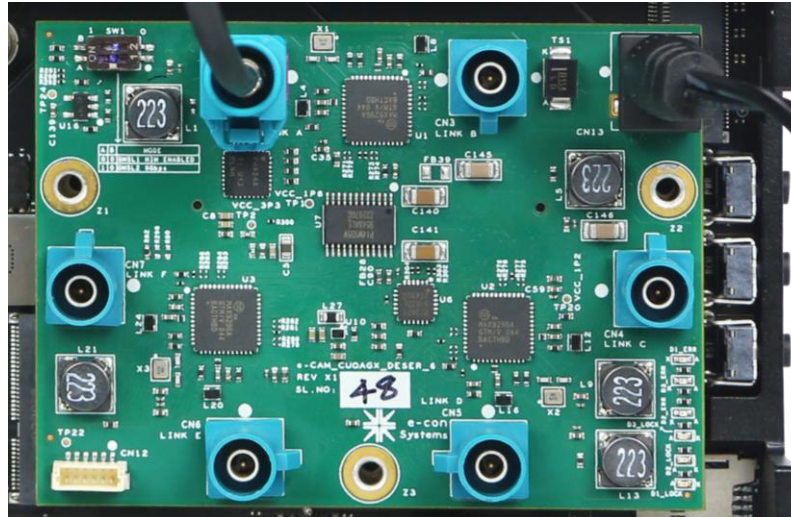


Figure 26: 12V Power Adapter connected to DC Jack of NileCAM81_CUOAGX Deserializer Board

With NileCAM81_CUOAGX connected to Jetson AGX Xavier™ development kit, the steps to power up and configure the Xavier™ setup are as follows:

8. Connect 19V, 3.42A supply to DC jack to power ON the Jetson Xavier™ development kit.
9. Press the power switch of Jetson development kit to boot the device.

The location of power switch on the Jetson AGX Xavier™ development kit is shown in following figure.

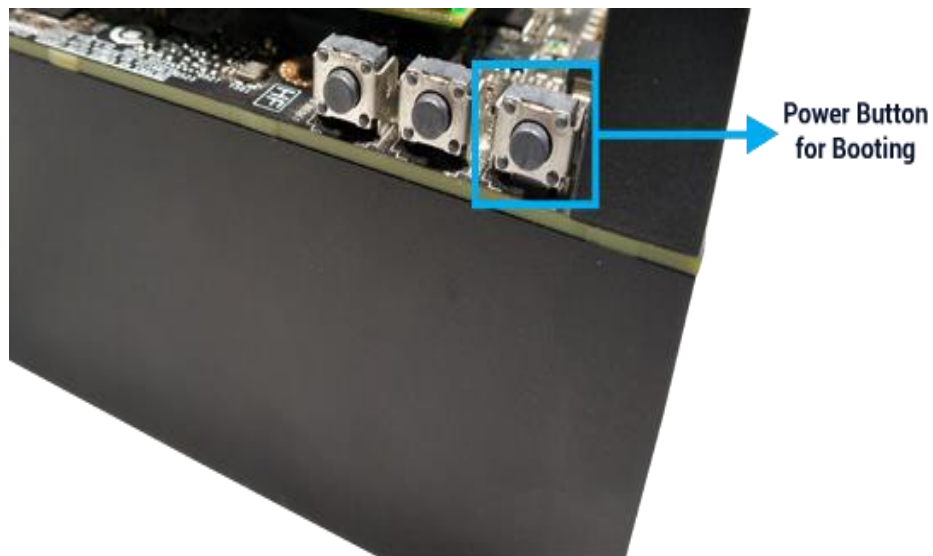


Figure 27: Location of Power Switch on Jetson AGX Xavier Development Kit

After powering ON the Jetson AGX Xavier™ development kit, the white color LED on Jetson AGX Xavier™ development kit will glow. This serves as an indication for power-up of base board as shown in following figure.

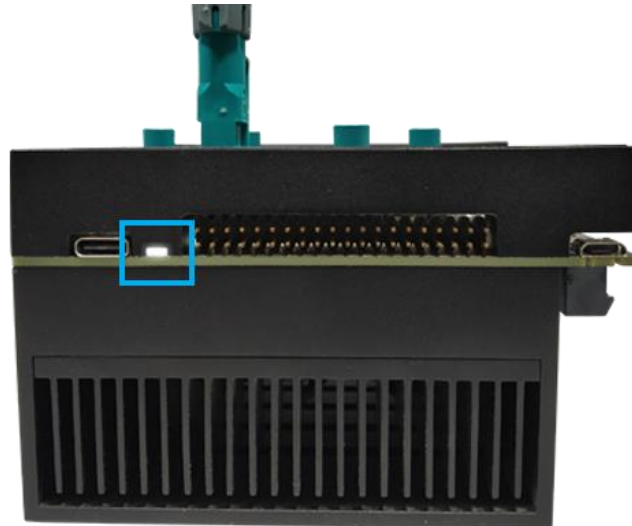


Figure 28: Power ON Indication White LED glowing on Jetson AGX Xavier Development Kit

Software Quick Setup

This section describes the extracting and flashing of e-con Systems provided binaries.

The commands and output messages in this manual are represented by different colors as listed in following table.

Table 5: Notation of Colors

Color	Notation
Blue	Commands running in Development PC
Red	Commands running in Jetson Board
Orange	Output messages in the Jetson Board

Before extracting and flashing the e-con provided binaries in Jetson Orin™/Xavier™ development kit, flash the Jetpack 5.1.2 provided by NVIDIA®, using the SDK manager.

The steps to flash the Jetpack are as follows:

1. Download the SDK manager using this <https://developer.nvidia.com/nvidia-sdk-manager> link.
2. Sign up to an account in NVIDIA® developer site to use the SDK manager.
3. Run the following command to install the SDK Manager in the host PC.

```
sudo apt-get install ./sdkmanager-  
[version].[build#].deb
```

After installing the SDK manager in the host PC, follow the instructions in the link <https://docs.nvidia.com/sdk-manager/install-with-sdcm-jetson/index.html>.

4. Copy the release package into the HOME directory of the flashed Jetson Orin™/Xavier™ development kit.

To extract and flash the binaries, follow these steps:

1. Run the following commands to extract the release package in the Jetson development kit to obtain the binaries.

```
tar -xaf e-CAM_YUV-GMSL-  
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_  
version>.tar.xz  
cd e-CAM_YUV-GMSL-  
PRODUCTS_JETSON_<L4T_version>_<release_date>_<release_  
version>
```

The package contains the necessary tools to immediately flash the binaries in Jetson Orin™/Xavier™ with the kernel, camera drivers, and application.

2. Run the following commands in the Jetson kit to install the binaries.

```
sudo chmod +x ./install_binaries.sh  
sudo -E ./install_binaries.sh 81
```

The Jetson reboots automatically after the installation of binaries.

The steps to be followed in the development board for launching the application are as follows:

1. Run the following command to check whether all the cameras are initialized properly.

```
$ sudo dmesg | grep -i 'Detected AR0821 sensor'
```

The output message appears as shown below.

```
Detected AR0821 sensor
```

The number of times the above message displayed corresponds to the number of cameras connected to the Jetson development kit.

2. Run the following command to check the presence of video nodes.

```
ls /dev/video*
```

The output message appears as shown below.

```
/dev/video*
```

Where (*) denotes the index number for each camera connected and detected in the Jetson development kit.

3. Run the following command to set the power mode to maximum for better performance.

```
sudo nvpmodel -m 0
```

4. Run the following Jetson clocks command before launching the ecam_tk1_guvcview application in Jetson development kit.

```
sudo jetson_clocks
```

To launch the ecam_tk1_guvcview application follow *e-CAM_TK1_GUVCView_Build_and_Install_Guide_<REV>.pdf* document.

After completing the procedure mentioned in above document, run below command to launch the application.

```
ecam_tk1_guvcview
```

When application is launched, you can view the screen similar to the screen shown below.


Image Controls


Video

User Controls:

Brightness: 0 - +

Contrast: 0 - +

Saturation: 16 - +

White Balance, Automatic: ☒

Gamma: 220 - +

Gain: 1 - +

Horizontal Flip: ☐

Vertical Flip: ☐

White Balance Temperature: 4500 - +

Sharpness: 2 - +

Camera Controls:

Exposure Auto: Full FOV Auto Mode ▼

Exposure Time, Absolute: 312 - +

ROI Window Size: 8 - +

HDR: Day HDR ▼

Frame Sync: Disable Frame Sync ▼

Denoise: 8 - +

Exposure Compensation: 33333 - +

Powerline Frequency: Auto ▼

Figure 29: Initial Window Appearance when Application is launched

Reference Documents

This section describes the software and hardware documents of NileCAM81_CUOAGX. You can download the software and hardware documents from [Developer Resources](#) website.

Software Documents

The software documents and its description are listed in following table.

S.NO	What I need	Documents to Refer
1	View the single camera stream and change the camera controls using GUI.	NileCAM81_CUOAGX_Linux_App_User_Manual_<REV>.pdf
2	Build and install the GUI based ecam_tk1_guvcview camera application.	e-CAM_TK1-GUVCView_Build_and_Install_Guide_<REV>.pdf
3	Stream and view all the cameras of NileCAM81_CUOAGX in Synchronous mode.	NileCAM81_CUOAGX_e-multicam_App_User_Manual_<REV>.pdf
4	Use Gstreamer to control the NileCAM81_CUOAGX camera on the Jetson development kit.	NileCAM81_CUOAGX_Gstreamer_Usage_Guide_<REV>.pdf
5	Build custom kernel with support for using NileCAM81_CUOAGX.	NileCAM81_CUOAGX_Developer_Guide_<REV>.pdf
6	Information about the directory structure and contents of the release package for NileCAM81_CUOAGX.	NileCAM81_CUOAGX-Release_Package_Manifest_<REV>.pdf
7	Integrate e-con's NileCAM81_CUOAGX with any host system other than what is supported by e-con Systems directly.	e-CAM82_CUMI0821_MOD_MCU_Protocol_AppNote_<REV>.pdf
8	Describes the working of Error Recovery Mechanism in NileCAM81_CUOAGX	E-CON_SYSTEMS_GMSL2_CAMERA_PRODUCTS_Error_Recovery_Guide_<REV>.pdf

Table 6: Description of Software Documents

Hardware Documents

The hardware documents and its description are listed in following table.

S.NO	Documents Name	Description
1	NileCAM81_CUOAGX_Datasheet_<REV>.pdf	Describes the features, connector pin-out details and mechanical dimensions of NileCAM81_CUOAGX.
2	NileCAM81_Datasheet_<REV>.pdf	Describes the features, connector pin-out details and mechanical dimensions of NileCAM81.
3	NileCAM81_CUOAGX_Lens_Datasheet_<REV>.pdf	Describes the optical specification of lenses used in NileCAM81_CUOAGX.
4	NILECAM81_1H01R1_3D.stp	3D file for NILECAM81_1H01R1
5	e-CAM82_CUMIO821_MOD_<REV>_TOP.dxf	2D file for e-CAM82_CUMIO821 RevX1 module top side.
6	e-CAM82_CUMIO821_MOD_<REV>_BOT.dxf	2D file for e-CAM82_CUMIO821 RevX1 module bottom side.
7	e-CAM82_CUMIO821_MOD_3D.stp	3D file for e-CAM82_CUMIO821_MOD

Table 7: Description of Hardware Documents

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

The systems are running in low-graphics mode error log appears on screen. How to solve this issue?

This error may occur in Jetson Orin™/Xavier™ when complete package is flashed. To solve this issue, reset the board.

Why the camera is not detected, even if the device is connected and powered ON?

This may occur on the deserializer board when it is not powered ON. Make sure that the deserializer board is powered ON, before the Jetson development kit.

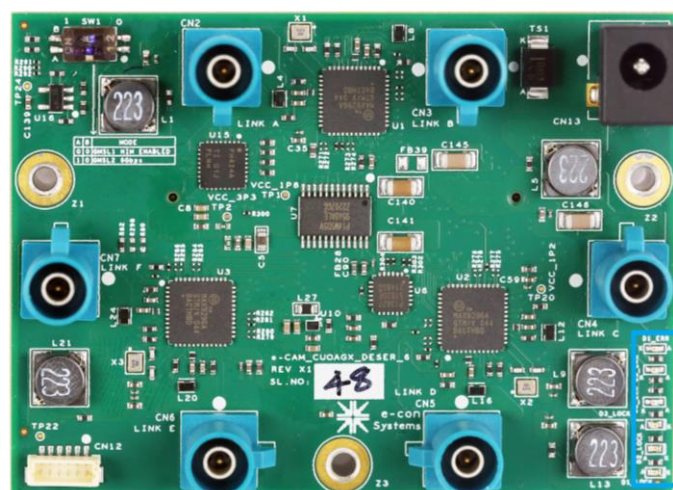
Why power ON indication LED of serializer is OFF, even if the device is connected?

When the power indicator LED of Serializer (D2) is OFF (with status LED of base board is ON), it means that power is not applied from base board. Check if the coaxial cable is inserted properly on either end and also check whether Power is connected properly to deserializer board CN13.

Why error LED of deserializer is ON, if device is connected?

Apart from Lock LED, the deserializer board has a Red error LED (D1). The LED glows, when the error in the GMSL data exceeds the set threshold. The glowing of Red LED does not affect the camera streaming.

The following figure shows the Error LED (Blue) on Deserializer board.



1. Does NileCAM81_CUOAGX camera work with any coaxial GMSL cable available in the market?

In general, any coaxial cable whose specifications equal to RTK031 cable can be used. If you need more details on different cable usage, please write to camerasolutions@e-consystems.com with your requirement.

2. What are the cable lengths that are supported in NileCAM81_CUOAGX?

Currently, e-con Systems provide a very flexible and robust 3m/15m coaxial GMSL cable along with this kit. For more lengths and customizations, please write to camerasolutions@e-consystems.com with your requirement.

3. Does NileCAM81_CUOAGX camera support OpenCV?

NileCAM81_CUOAGX works using V4L2 APIs and is V4L2 compliant. So, any V4L2 based application can be used to access this camera. OpenCV is also compatible since it uses V4L2 to access the camera.

Please refer to https://www.e-consystems.com/Articles/Camera/accessing_cameras_in_opencv_with_high_performance.asp for detailed information about OpenCV support in e-con Systems cameras.

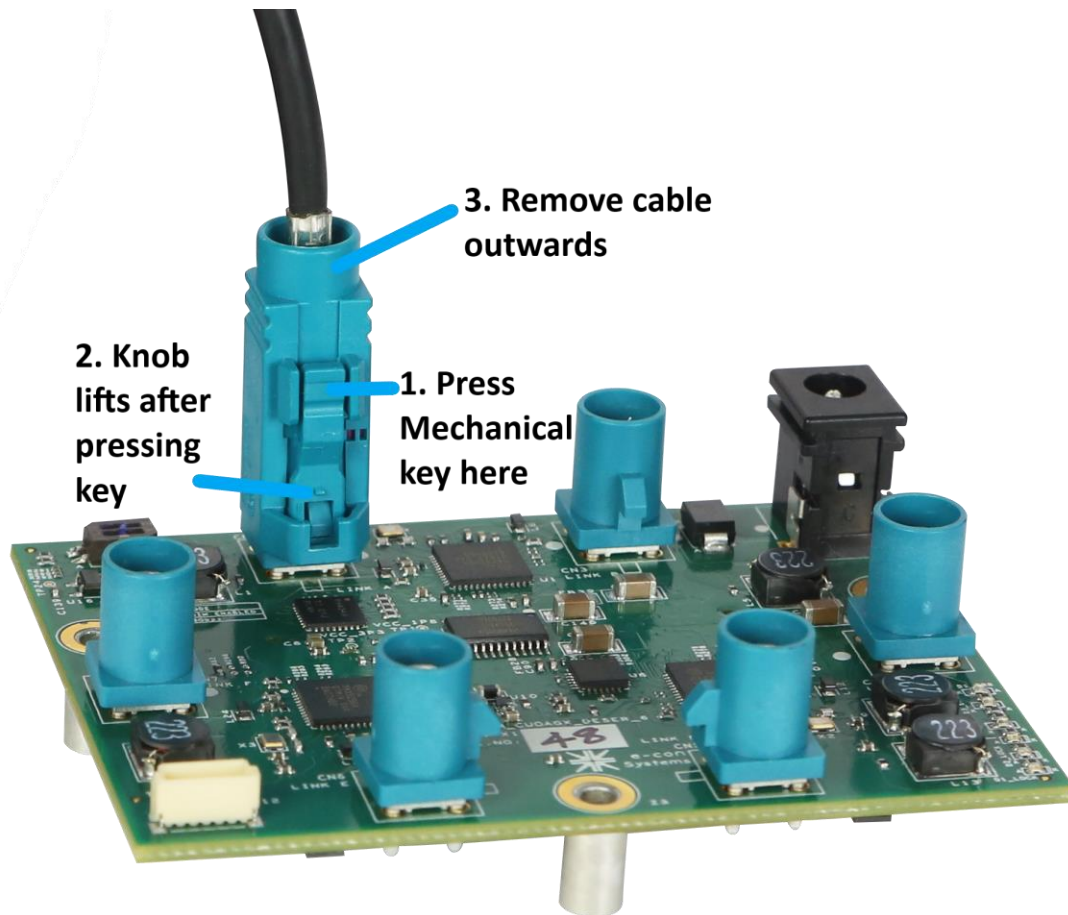
4. How can I get the updated package?

Please login to the [Developer Resources](#) website and download the latest release package.

5. How to remove the coaxial cable from FAKRA plug?

Care needs to be taken while removing the coaxial cable from the PCB. The FAKRA jack on coaxial cable locks with mechanical key of the FAKRA plug on board. The lock between plug and jack needs to be released, to remove the cable. On pressing the jack at the specified location below, the knob lifts and the cable can be removed.

The following figure shows the stages of removing FAKRA jack (coaxial cable) from PCB.



6. Why the mounting hole screws are not fitting (Deserializer to Xavier™) in NileCAM81_CUOAGX?

Xavier™ does not have threads in their kit, so screws will not fit in it.

What's Next?

After understanding the specifications of camera daughter board and instructions on how to use this daughter board with Jetson Orin™/Xavier™ development kit, you can refer to the following documents to understand more about NileCAM81_CUOAGX.

- *NileCAM81_CUOAGX Developer Guide*
- *NileCAM81_CUOAGX Linux App User Manual*

Glossary

API: Application Program Interface.

CMOS: Complementary Metal Oxide Semiconductor.

GMSL: Gigabit Multimedia Serial Link.

GUI: Graphical User Interface.

HDMI: High-Definition Multimedia Interface.

ISP: Image Signal Processor.

LED: Light-Emitting Diode.

MCU: Micro Controller Unit.

PCB: Printed Circuit Board.

V4L2: Video4Linux2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Contact Us

If you need any support on NileCAM81_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	01-Aug-2022	Initial Draft	Camera Products team
1.1	09-Nov-2022	Minor changes with support for Jetpack 5.0.2	Camera Products team
1.2	23-Jan-2023	Updated Sensor format 1/2 to 1/1.7	Camera Products team
1.3	30-Jan-2023	Updated Images	Camera Products team
1.4	10-Feb-2023	Added Sync Support	Camera Products Team
1.5	22-Mar-2023	Updated to L4T35.2.1	Camera Products Team
1.6	24-May-2023	Updated to L4T35.3.1	Camera Products Team
1.7	07-Jun-2023	Updated Images	Camera Products Team
1.8	19-Oct-2023	Updated to L4T35.4.1	Camera Products Team
1.9	08-Jan-2024	Renamed the release package and added error recovery and health monitoring support.	Camera Products Team